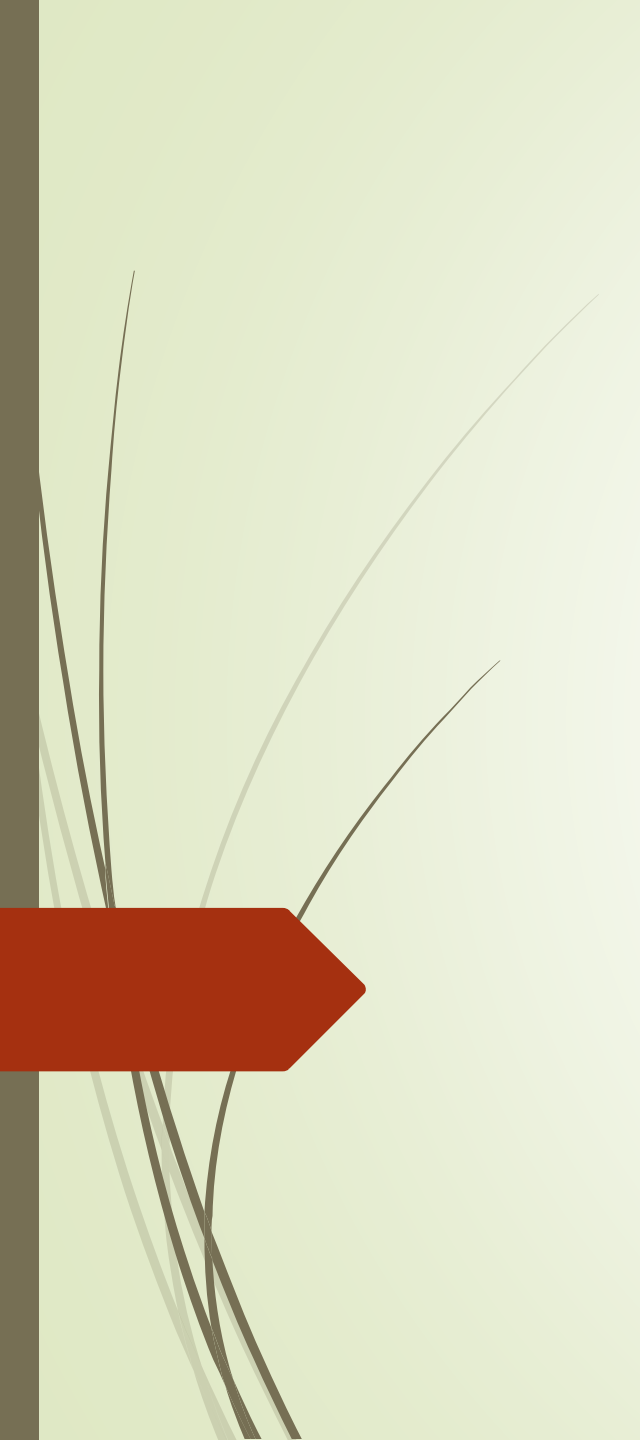




Natural Language Processing



Text Similarity

- Text Similarity is **the process of comparing one piece of text with another and finding the similarity between them.**
- Semantic Textual Similarity is **the task of evaluating how similar two texts are in terms of meaning.**
- Example:
- T1: The dog bites a man
- T2: The man was bitten by a dog

Text Similarity

- Source: a dog bites a man
- Suspicious: a man was bitten by a dog
- **Generate sets of unigrams**
- source_unigram
 - {**The, dog, bites, a, man**}
- suspicious_unigrams
 - {**The, man, was, bitten, by, a, dog**}
- **Compute Similarity**
- $\text{Sim} = 4/\min(5,7) = 4/5 = 0.8$

Text Similarity Methods

Jaccard Similarity Coefficient (JSC)

Overlap Similarity Coefficient (OSC)

Dice Similarity Coefficient (DSC)

Euclidean Distance Measure (EDM)

Cosine Similarity Coefficient (CSC)

$$JSC(T_1, T_2) = \frac{|T_1 \cap T_2|}{|T_1 \cup T_2|}$$

$$OSC(T_1, T_2) = \frac{|T_1 \cap T_2|}{\min(|T_1|, |T_2|)}$$

$$DSC(T_1, T_2) = \frac{2|T_1 \cap T_2|}{|T_1| + |T_2|}$$

$$EDM(T_1, T_2) = \sqrt{\sum_{i=1}^n (T_{1i} - T_{2i})^2}$$

$$CSC(T_1, T_2) = \frac{T_1 \cdot T_2}{|T_1|^2 \cdot |T_2|^2} = \frac{\sum_{i=1}^n (T_{1i} \cdot T_{2i})}{\sqrt{\sum_{i=1}^n T_{1i}^2} \sqrt{\sum_{i=1}^n T_{2i}^2}}$$

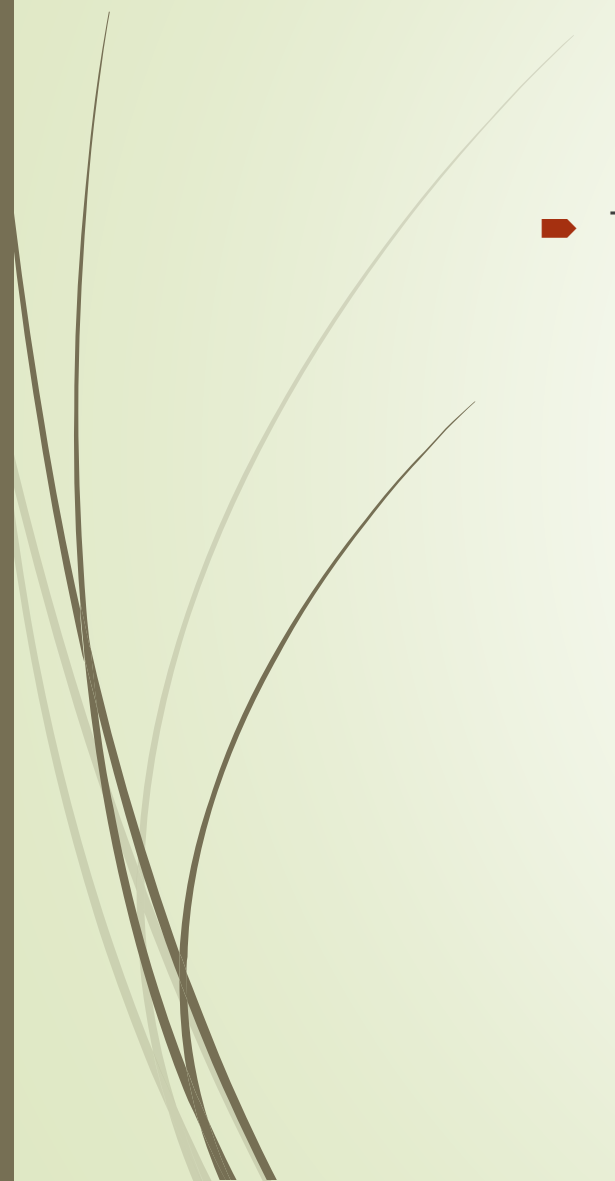
Text to Numeric

T1: He is going to school

T2: She is going to university

Compute Vocabulary: {He, is, going, to, school, she, university}

Token	T1	T2
He	1	0
Is	1	1
Going	1	1
To	1	1
School	1	0
She	0	1
university	0	1



▀ Thanks